Quiz 1: 17/02/25 Quiz 2: 1/04/25 endSem: 9/05/25

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PH1020 Physics II
Syllabus 2025

Learning objectives:

To enable the students to:

1) describe electromagnetic phenomena using the language of vector calculus.

2) determine electric and magnetic fields arising from simple configurations of static charges as well as steady currents in vacuum and in matter.

3) solve problems involving the propagation of electromagnetic waves in vacuum and in matter.

Quiz 1, Quiz 2:

3MCQ: 3x2 = 6
Box : 6x1 = 6
2 long : 2x4 = 8

Text Book:

Introduction to Electrodynamics
by *David J. Griffiths*
4th Edition (South Asia Edition)

Detailed Syllabus with section numbers from Griffith 4rd Edition:

Unit 1: Electrostatics & Electric Fields in Matter

[2.1] The Electric Field, [2.1.2] Coulomb's Law, [2.1.3] The Electric Field, [2.1.4] Continuous Charge Distributions, [2.2] Divergence and Curl of Electrostatic Fields, [2.2.1] Field Lines, Flux, and Gauss's Law, [2.2.3] Applications of Gauss's Law, [1.5] The Dirac Delta Function, [2.3] Electric Potential, [2.3.4] The Potential of a Localized Charge, Distribution, [2.3.5] Boundary Conditions, [2.4] Work and Energy in Electrostatics, [2.4.1] The Work It Takes to Move a Charge, [2.4.2] The Energy of a Point Charge Distribution, [2.4.3] The Energy of a Continuous Charge Distribution, [2.5] Conductors, [2.5.1] Basic Properties, [2.5.2] Induced Charges, [2.5.3] Surface Charge and the Force on a Conductor, [2.5.4] Capacitors, [2.3.3] Poisson's Equation and Laplace's Equation, [3.1.2] Laplace's Equation in One Dimension, [3.4] Multipole Expansion, [3.4.2] The Monopole and Dipole Terms, [LN1]*Quadrupole moment tensor*, [3.4.4] The Electric Field of a Dipole, [4.1.3] Force and torque on a dipole due to an external static electric field, [LN2] *Energy of an ideal dipole in an electric field, The interaction energy between two dipoles*.

Electric Fields in Matter: [4.1] Polarization, [4.1.1] Dielectrics, [4.1.2] Induced Dipoles, [4.2] The Field of a Polarized Object, [4.2.1] Bound Charges, [4.3] The Electric Displacement, [4.3.1] Gauss's Law in the Presence of Dielectrics, [4.3.3] Boundary Conditions, [4.4] Linear Dielectrics, [4.4.1] Susceptibility, Permittivity, Dielectric Constant, [4.4.3] Energy in Dielectric Systems

Unit 2: Magnetostatics & Magnetic Fields in Matter

[5.1] The Lorentz Force Law, [5.1.1] Magnetic Fields, [5.1.2] Magnetic Forces, [5.1.3] Currents, [5.2] The Biot-Savart Law, [5.2.1] Steady Currents, [5.2.2] The Magnetic Field of a Steady Current, [5.3.1] Straight-Line Currents, [5.3.2] The Divergence and Curl of B, [5.3.3] Ampère's Law, [5.4] Magnetic Vector Potential, [5.4.1] The Vector Potential, [5.4.2] Boundary Conditions, [5.4.3] Multipole Expansion of the Vector Potential.

Magnetic Fields in Matter: [6.1] Magnetization, [6.1.1] Diamagnets, Paramagnets, Ferromagnets, [6.1.2] Torques and Forces on Magnetic Dipoles, [6.2] The Field of a Magnetized Object, [6.2.1] Bound Currents, [6.3] The Auxiliary Field H, [6.3.1] Ampère's Law in Magnetized Materials, [6.3.3] Boundary Conditions, [6.4] Linear and Nonlinear Media, [6.4.1] Magnetic Susceptibility and Permeability.

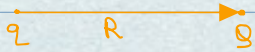
Unit 3: Electrodynamics & Electromagnetic Waves

[7.1] Electromotive Force, [7.1.1] Ohm's Law, [7.1.2] Electromotive Force, [7.1.3] Motional emf, [7.2] Electromagnetic Induction, [7.2.1] Faraday's Law, [7.2.2] The Induced Electric Field, [7.2.3] Inductance, [7.2.4] Energy in Magnetic Fields, [7.3.1] Electrodynamics Before Maxwell, [7.3.2] How Maxwell Fixed Ampère's Law [7.3.3] Maxwell's Equations, [7.3.5] Maxwell's Equations in Matter, [7.3.6 Boundary Conditions, [10.1, LN3] *Gauge invariance-Coulomb and Lorentz gauge*, [8.1.1] The Continuity Equation, [8.1.2] Poynting's Theorem, [8.2.4] Angular Momentum.

Electromagnetic Waves: [9.2] Electromagnetic Waves in Vacuum, [9.2.1] The Wave Equation for E and B, [9.2.2] Monochromatic Plane Waves, [9.2.3] Energy and Momentum in Electromagnetic Waves, [9.3] Electromagnetic Waves in Matter, [9.3.1] Propagation in Linear Media, [9.3.2] Reflection and Transmission at Normal Incidence, [9.3.3] Reflection and Transmission at Oblique Incidence (Law of reflection, law of refraction—Snell's law, Fresnel's equations, Brewster's angle)

**LN = Lecture Notes*

Coulomb's Law: (based on experiments)



$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2}\hat{e}_r = Q\vec{E} \qquad (Q \rightarrow \text{test charge})$$

$\varepsilon_0 \rightarrow$ permitivity of free space $= 8.85 \times 10^{-12}\ \frac{C^2}{Nm^2}$

Superposition principle: states that the interaction between any two charges is completely unaffected by the presence of others

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \ldots + \vec{F}_n$$

Notations:

$q_i \rightarrow i^{th}$ source charge

$Q \rightarrow$ test charge

$\vec{r}$ (or $\vec{R}$) $\rightarrow$ Position vector of Q / Point of interest

$\vec{r}_i' \rightarrow$ position vector of q_i

$\vec{r}_{qQ} \rightarrow$ vector from q to Q

$\vec{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z$

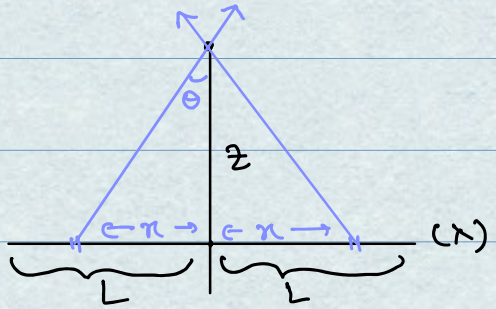
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\left(\frac{a_i}{r_{aiQ}^2}\hat{\gamma}_{ai9}\right) = \vec{E}_1 + \vec{E}_2 + \ldots + \vec{E}_n$$

Continuous charge distributions:

- Line charge: (λ) $\quad \vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda\, dl'}{|\vec{r}-\vec{r}'|^3}(\vec{r}-\vec{r}')$

- Surface charge: (σ) $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma\, da'}{|\vec{r}-\vec{r}'|^3}(\vec{r}-\vec{r}')$

- Volume charge: (ρ) $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho\, dv'}{|\vec{r}-\vec{r}'|^3}(\vec{r}-\vec{r}')$

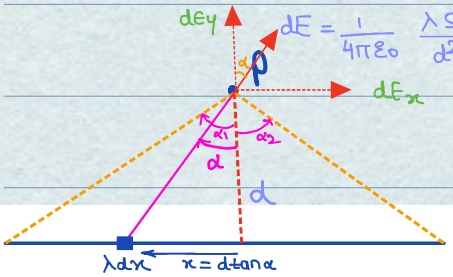


$$d\vec{E} = 2\times \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{(x^2+z^2)}\cos\theta$$

$$\cos\theta = \frac{z}{\sqrt{x^2+z^2}}$$

Derivations:

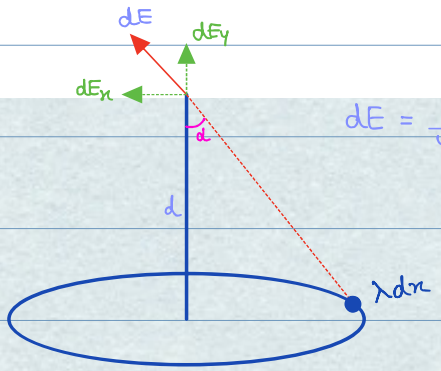
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$$E_x = \int_{-\alpha_2}^{\alpha_1} \frac{\lambda}{4\pi\varepsilon_0}\sin\alpha\, d\alpha = \frac{\lambda}{4\pi\varepsilon_0}(\cos\alpha_2 - \cos\alpha_1)$$

$$E_y = \int_{-\alpha_2}^{\alpha_1} \frac{\lambda}{4\pi\varepsilon_0}\cos\alpha\, d\alpha = \frac{\lambda}{4\pi\varepsilon_0}(\sin\alpha_1 + \sin\alpha_2)$$

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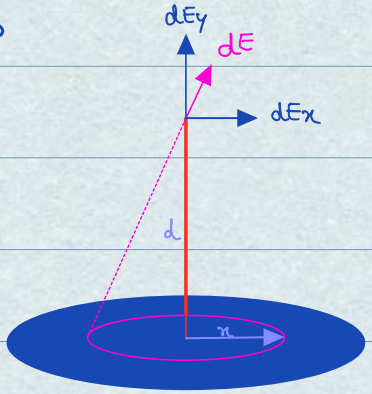
$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{(r^2+d^2)}$$

$$E_x = 0$$

$$E_y = \int_0^{2\pi} \frac{1}{4\pi\varepsilon_0}\frac{\lambda r\, d\theta}{(r^2+d^2)}\frac{d}{(r^2+d^2)^{1/2}} = \frac{\lambda r d}{2\varepsilon_0 (r^2+d^2)^{3/2}}$$

$d \gg R \Rightarrow$ Similar to point charge $\frac{Q}{4\pi\varepsilon_0 d^2}$

$(Q = 2\pi r\lambda)$



$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot 2\pi x\,dx}{(d^2+x^2)}$$

$$E_x = 0$$

$$E_y = \int_0^R \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot 2\pi x\,dx}{(d^2+x^2)} \cdot \frac{d}{(d^2+x^2)^{1/2}}$$

$$E_y = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{d}{\sqrt{d^2+R^2}}\right]$$

Electrostatic Potential

- Electrostatic field: conservative vector, derived from central force
- $\vec{E}(\vec{r}) = -\vec{\nabla}\phi(\vec{r})$ conservative field can be expressed as gradient of scalar potential

$$\vec{\nabla} \times \vec{E}(\vec{r}) = 0$$

↓ should be static ; if not then not true

- Potential is irrotational in region where it's defined. Curl is zero

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{|\vec{r} - \vec{r}'|} + C$$

- For a system of charges,

$$\phi(\vec{\gamma}) = \sum_{i=1}^{n} \frac{1}{4\pi\varepsilon_0}\frac{q_i}{|\vec{\gamma} - \vec{\gamma}_i'|} + C$$

Dirac Delta function:

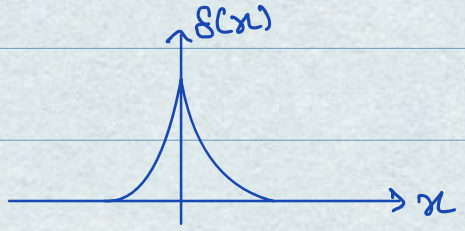
Consider the vector function $V = \frac{1}{r^2}\hat{r}$

$$\nabla \cdot V = \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{1}{r^2}\right) = 0 \quad \text{(divergence)}$$

But when $r=0$, $\nabla \cdot V$ blows up → vanishes at origin

1D dirac delta function

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$



- $\int_{-\infty}^{\infty} \delta(x)\,dx = 1$

- $\delta(x) \cdot f(x) = 0 \quad (x \neq 0)$ $f(x)$ is some function

- $\int_{-\infty}^{\infty} f(x)\,\delta(x) = f(0)\int_{-\infty}^{\infty} \delta(x)\,dx = f(0)$

- $\delta(x-a) = \begin{cases} 0 & x \neq a \\ \infty & x = a \end{cases} \Rightarrow \int_{-\infty}^{\infty} f(x)\,\delta(x-a) = f(a)$

3D delta dirac function: $\delta^3(\vec{r}) = \delta(x)\,\delta(y)\,\delta(z)$

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

Field lines, Flux, Gauss Law

- Field lines never cross each other
- Field lines that originate from a +ve charge must pass through the surface or terminate at a negative charge inside

- Flux: number of field lines per unit area (⊥ to surface)

$$Flux = \int_S F\cos\theta \cdot da = \int_S \vec{F}.\hat{(n)}da = \int_S \vec{F}\cdot\vec{da}$$

- For a point charge q, flux through a sphere of radius r is

$$\oint \vec{E}\cdot d\vec{A} = \int \left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{e}_r\right)\left(r^2 \sin\theta\, d\theta\, d\phi\, \hat{e}_r\right) = \frac{q}{\varepsilon_0}$$

($\int \sin\theta\, d\theta = 2$, $\int d\phi = 2\pi$)

- Gauss law: $\oint_S \vec{E}\cdot d\vec{a} = \frac{Qenc}{\varepsilon_0}$

Qenc → total charge enclosed within the surface

- using divergence theorm, $\oint_S \vec{E}\cdot d\vec{A} = \int_V \left(\vec{\nabla}\cdot\vec{E}\right) d\tau$

$$Qenc = \int_V \rho\, d\tau$$

$\Rightarrow$ Gauss law becomes : $\int_V \left(\vec{\nabla}\cdot\vec{E}\right) d\tau = \int_V \left(\frac{\rho}{\varepsilon_0}\right) d\tau$

This holds for any volume

$\Rightarrow \vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$: Gauss law in differential form

ρ: charge per unit volume

- Field outside uniformly charged solid sphere of radius r and total charge q:

Gaussion surface of radius r ($r>R$)

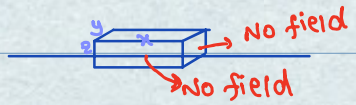
$$\int_S |\vec{E}|\, da = |\vec{E}| \int_S da = |\vec{E}| \cdot 4\pi r^2 = \frac{q}{\varepsilon_0} \Rightarrow \vec{E} = \frac{q}{4\pi\varepsilon_0 r^2}\hat{e}_r$$

(since E is uniform in direction of $\hat{e}_r$)

- Spherical Symmetry
- cylindrical symmetry
- Plane Symmetry

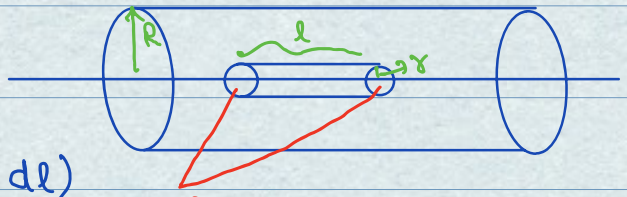
- infinite sheet with charge density σ, find $\vec{E}$ at distance z from plane of sheet

$q_{enc} = \sigma \times xy = \sigma xy$



$$\int_S \vec{E}.\vec{dA} = |\vec{E}| \int_S da = |\vec{E}|\, 2xy = \frac{q_{enc}}{\varepsilon_0} = \frac{\sigma xy}{\varepsilon_0} \Rightarrow |E| = \frac{\sigma}{2\varepsilon_0}$$

- Long cylinder carries charge density proportional to the distance from axis. Find field inside cylinder.



$$Q_{enc} = \int \rho \, d\tau$$

$$= \int_0^r (kz)(z.d\phi.dz.dl)$$

$$= k \times 2\pi \times l \times \frac{r^3}{3} = \frac{2}{3}\pi k l r^3$$

$$\int_S \vec{E}.\vec{da} = \int_S |E|.da = |E| \int_S da = |E| 2\pi r l = \frac{Q_{enc}}{\varepsilon_0} = \frac{2\pi}{3\varepsilon_0} k l r^3$$

$$\Rightarrow |E| = \frac{1}{3\varepsilon_0} k r^2$$

Work:

- Electrostatic potential can be defined as the work done in bringing a unit charge (+ve) from infinity to that point.
- Work done for assembly of discrete charges:

$$W = \frac{1}{2} \sum_i^N \sum_{j<i} q_i \phi j = \frac{1}{4\pi\varepsilon_0} \cdot \frac{1}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ (j \neq i)}}^N \frac{q_i q_j}{|\vec{r}_i - \vec{r}_j|}$$

- Work done for:

  line charge: $\frac{1}{2}\int \phi \lambda dl \rightarrow \phi(\bar{r}) = \frac{1}{4\pi\varepsilon_0}\int_L \frac{\lambda dl_1}{|\bar{r}-\bar{r}'|} + C$

  Surface charge: $\frac{1}{2}\int \phi \sigma da \rightarrow \phi(\bar{r}) = \frac{1}{4\pi\varepsilon_0}\int_A \frac{\sigma da'}{|\bar{r}-\bar{r}'|} + C$

  Volume charge: $\frac{1}{2}\int \phi \rho dv \rightarrow \phi(\bar{r}) = \frac{1}{4\pi\varepsilon_0}\int_v \frac{\rho dv'}{|\bar{r}-\bar{r}'|} + C$

- $\vec{E} = -\vec{\nabla}\phi$

$\vec{\nabla} \times \vec{E} = \vec{0}$

http://www.csupomona.edu/~ajm/materials/delsph.pdf

http://www.csupomona.edu/~ajm/materials/delcyl.pdf

If we know V, E can easily be found.

$\frac{dE_x}{dy} = \frac{dE_y}{dx}$; $\frac{dE_y}{dz} = \frac{dE_z}{dy}$; $\frac{dE_z}{dx} = \frac{dE_x}{dz}$

$$\vec{E} = \hat{i}\frac{d\phi}{dx} + \hat{j}\frac{d\phi}{dy} + \hat{k}\frac{\partial\phi}{\partial z}$$

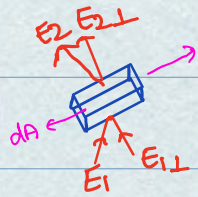
$\vec{\nabla}.\vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \nabla^2\phi = \frac{\rho}{\varepsilon_0}$. $\rightarrow$ Poisson's equation

In a charge-free region; $\nabla^2\phi = 0 \rightarrow$ Laplace equation

Electrostatic boundary charges:

when electric field passes through a surface charge, it suffers a discontinuity

$$\int_{\text{pill box}} E\,dA = \frac{q_{enc}}{\varepsilon_0} = \frac{\sigma\,dA}{\varepsilon_0}$$



small pill box, half above surface, half below, thickness ε

$$(E_{2\perp} - E_{1\perp})\,\delta A = \frac{\sigma\,dA}{\varepsilon_0} \Rightarrow E_{2\perp} - E_{1\perp} = \frac{\sigma}{\varepsilon_0}$$ → discontinuous at surface

changes from $-\frac{\sigma}{2\varepsilon_0}\hat{n}$ to $\frac{\sigma}{2\varepsilon_0}\hat{n}$ at the boundary

parallel to the surface: $E_{1\parallel} = E_{2\parallel}$ → continuous

work done in assembling charges:

$$W = \frac{1}{2}\sum_i \sum_{j<i} q_i \phi_j = \frac{1}{4\pi\varepsilon_0}\cdot\frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j=1 \\ j\neq i}}^{N}\frac{q_i q_j}{|\vec{r}_i - \vec{r}_j|}$$

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow W = \frac{\varepsilon_0}{2}\int \phi\,\vec{\nabla}\cdot\vec{E}\,dV$$

$$\vec{\nabla}\cdot(\phi\vec{E}) = \vec{\nabla}\phi\vec{E} + \phi\vec{\nabla}\cdot\vec{E}$$

$$\int_S \vec{E}\cdot d\vec{s} = \int_V (\vec{\nabla}\cdot\vec{E})\,dV$$

$$W = \frac{\varepsilon_0}{2}\left[\oint \phi\vec{E}\cdot d\vec{a} - \int \left(\vec{\nabla}\phi\vec{E}\right) dV\right]$$

$$W = \frac{\varepsilon_0}{2}\left[\oint \phi\vec{E}\cdot d\vec{a} + \int \left|\vec{E}\right|^2 dV\right]$$

$$W = \frac{\varepsilon_0}{2} \int_{\text{all space}} \left|\vec{E}\right|^2 dV \longrightarrow \text{energy stored always +ve}$$

Energy of a +ve and -ve charge seperated by a distance r is

$$W = \frac{1}{4\pi\varepsilon_0} \frac{q(-q)}{r^2} = \frac{-q^2}{4\pi\varepsilon_0 r^2} \quad (-ve)$$

It is doing work for us

Energy stored in Electric field of a point charge

$$W = \frac{\varepsilon_0}{2} \left(\frac{1}{4\pi\varepsilon_0}\right)^2 \int \frac{q^2}{r^4} (r^2 \sin\theta \, dr \, d\theta \, d\phi) = \frac{q^2}{8\pi\varepsilon_0} \int_0^\infty \frac{dr}{r^2} = \infty$$

For a spherical shell, charge is confined to the surface, but the field is present everywhere outside the surface

$\frac{1}{2}\varepsilon_0 \left|\vec{E}\right|^2 \longrightarrow$ Energy per unit volume (stored in field)

$\frac{\rho\phi}{2} \longrightarrow$ Energy per unit volume (stored in charges)

Superposition of Energy:

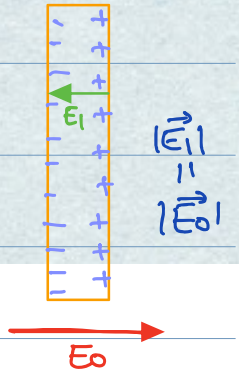
$$W = \frac{\varepsilon_0}{2}\int(\vec{E}_1+\vec{E}_2)^2 dV = \frac{\varepsilon_0}{2}\int(E_1^2+E_2^2+2\vec{E}_1\cdot\vec{E}_2)\,dV$$

$$\Rightarrow W_{total} = W_1 + W_2 + \varepsilon_0 \int (\vec{E}_1 \cdot \vec{E}_2) dV$$

Conductor:

1) $E = 0$ inside conductor

If we keep a conductor in an external field E_0, the free charges move in respective directions create induced charges, produce E_1 opposite to E_0 (instantaneous process)



2) $\rho = 0$ inside a conductor

Charges still exist, but equal amount of + and − charges

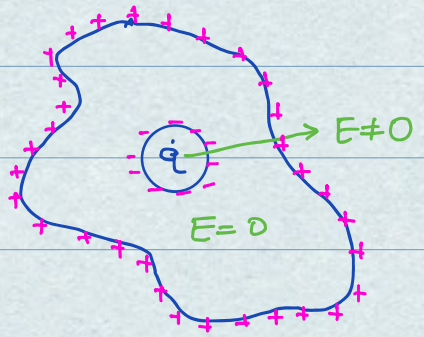
3) Any net charge resides on the surface

4) A conductor is an equipotential surface

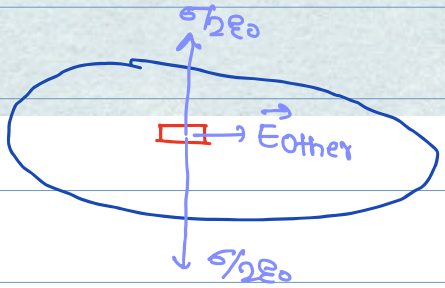
$V(a) - V(b) = -\int_b^a \vec{E}\cdot d\vec{l} = 0$ ($E = 0$)

5) E is perpendicular to the surface just outside the container

Field inside cavity in conductor is not zero



Boundary condition implies that field immediately outside $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$



$\vec{E}_{\text{at patch}} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} + \vec{E}_{\text{other}}$

$\Rightarrow \vec{E} = \vec{E}_{\text{other}}$

$P = \frac{\vec{E}}{A} = \sigma\vec{E}$

$\vec{E}_{\text{above}} = \vec{E}_{\text{other}} + \frac{\sigma}{2\varepsilon_0}\hat{n}$

$\vec{E}_{\text{below}} = \vec{E}_{\text{other}} - \frac{\sigma}{2\varepsilon_0}\hat{n}$

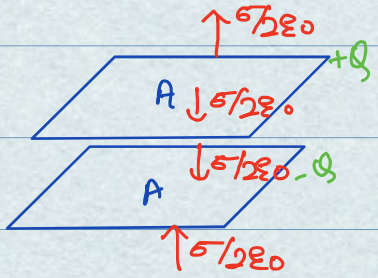
$\Rightarrow \vec{E}_{\text{other}} = \frac{1}{2}(\vec{E}_{\text{above}} + \vec{E}_{\text{below}}) = \vec{E}_{\text{avg}}$

for conductor, field inside is 0 and outside is $\frac{\sigma}{\varepsilon_0}\hat{n} \Rightarrow E_{\text{avg}} = \frac{\sigma}{2\varepsilon_0}\hat{n}$

$\Rightarrow \vec{P} = \frac{\sigma^2}{2\varepsilon_0}\hat{n}$ outward electrostatic pressure on the surface

Capacitors: $E \propto Q$, $V \propto Q \Rightarrow E \propto V$

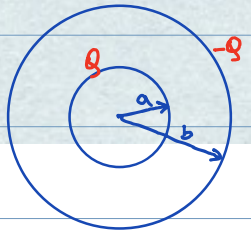
$\Rightarrow C = \frac{Q}{V}$ unit: Coulombs per volt / farad



$\sigma = \frac{Q}{A}$

$\Rightarrow E = \frac{\sigma}{\varepsilon_0}$ inside capacitor

$V = \frac{Q}{A\varepsilon_0} d \Rightarrow C = \frac{\varepsilon_0 A}{d}$



$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{e_r}$

$V = \int_a^b E \cdot dL = \frac{Q}{4\pi\varepsilon_0}\int_a^b \frac{dr}{r^2} = \frac{Q}{4\pi\varepsilon_0}\frac{b-a}{ab}$

$\Rightarrow C = 4\pi\varepsilon_0 \frac{ab}{b-a}$

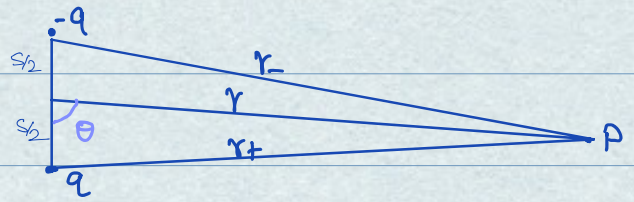
Work done to charge a capacitor

q be charge on +ve plate at some instant,

$V = q/C$

$dw = dq.V = \frac{q\,dq}{C} \Rightarrow w = \frac{q^2}{2C} = \frac{1}{2}CV^2$

Dipoles:



$$V_P = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r_+} - \frac{q}{r_-}\right]$$

$$r_\pm^2 = r^2 + \frac{s^2}{4} \mp rs\cos\theta = r^2\left[1 \mp \frac{s}{r}\cos\theta + \frac{s^2}{4r^2}\right]$$

$$\frac{1}{r_+} = \frac{1}{r}\left[1 \mp \frac{s}{r}\cos\theta\right]^{-1/2} = \frac{1}{r}\left[1 \pm \frac{s}{2r}\cos\theta\right]$$

$$\frac{1}{r_-} = \frac{1}{r}\left[1 \mp \frac{s}{2r}\cos\theta\right] \qquad r >>> s$$

$$\frac{1}{r_+} - \frac{1}{r_-} = \frac{s}{r^2}\cos\theta \Rightarrow V(P) = \frac{q}{4\pi\varepsilon_0}\frac{s\cos\theta}{r^2}$$

$q \cdot s = \vec{p}$ (dipole moment)

V (monopole): $\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ V (dipole): $\frac{|\vec{p}|\cos\theta}{4\pi\varepsilon_0 r^2}$

$\vec{E}$ vary as $\frac{1}{r^3}$

Multipole expansion of scalar potential

$$V(\vec{r}) = \sum_{i=1}^{n} \frac{q_i}{4\pi\varepsilon_0 R_i} \qquad R_i = |\vec{r} - \vec{r_i}|$$

$$= (r^2 + r_i^2 - 2rr_i\cos\theta)^{1/2}$$

$$= r(1+t)^{1/2}$$

$$t = -\frac{2r_i}{r}\cos\theta_i + \left(\frac{r_i}{r}\right)^2$$

$$(1+t)^{-1/2} = 1 - \frac{t}{2} + \frac{3t^2}{8} \cdots$$

$$\frac{1}{R_i} = \frac{1}{r}(1+t)^{-1/2} = \frac{1}{r} + \frac{r_i}{r^2}\cos\theta_i + \frac{1}{2}\frac{r_i^2}{r^3}(3\cos^2\theta_i - 1)$$

monopole — dipole — $\frac{\vec{p}\cdot\hat{r}}{4\pi\varepsilon_0 r^2}$

$$\Rightarrow V(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r}\sum_{i=1}^{n} q_i + \frac{1}{4\pi\varepsilon_0 r^2}\sum_{i=1}^{n} q_i r_i\cos\theta_i +$$

monopole moment — dipole moment $\vec{p}\cdot\hat{r} = \sum_{i=1}^{n} q_i r_i\cos\theta$

$$\frac{1}{4\pi\varepsilon_0 r^3}\sum_{i=1}^{n}\frac{q_i r_i^2}{2}(3\cos^2\theta_i - 1) + \cdots$$

quadrupole

$$V(\vec{r}) = V_m(\vec{r}) + V_d(\vec{r}) + V_q(\vec{r}) + \cdots$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{l=0}^{\infty}\frac{1}{r^{l+1}}\left[\sum_{i=1}^{n} q_i r_i^l P_l(\cos\theta_i)\right]$$

$$\cos\theta_i = \frac{l_x x_i + l_y y_i + l_z z_i}{r_i}$$

Monopole moment is a unique property. But

dipole and quadrupole terms depend on the choice of origin, since they depend on the absolute value of the origin.

$$\vec{p_0} = \vec{p} - (\sum q_i)\vec{a} \quad (a \rightarrow \text{shifted origin})$$

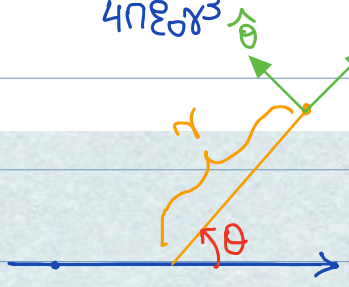
Field due to dipole:

$$V(\vec{r}) = \frac{\vec{p}\cdot\hat{r}}{4\pi\varepsilon_0 r^2} = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}$$

$$E = -\nabla V = -\frac{\partial V}{\partial r}\hat{r} - \frac{\partial V}{r\partial\theta}\hat{\theta} - \frac{\partial V}{r\sin\theta\,\partial\phi}\hat{\phi}$$

$$= \frac{2p\cos\theta}{4\pi\varepsilon_0 r^3}\hat{r} + \frac{p\sin\theta}{4\pi\varepsilon_0 r^3}\hat{\theta} + 0\,\hat{\phi}$$

$$\vec{E_d}(r,\theta) = \frac{p}{4\pi\varepsilon_0 r^3}\left[2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\theta\right]$$



coordinate free form: $\frac{1}{4\pi\varepsilon_0 r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$

$$\vec{p} = |p|\cos\theta + |p|\sin\theta = p\cos\theta\,\hat{r} - p\sin\theta\,\hat{\theta}$$

Solution to Laplace's equation

In charge free region: $\nabla^2 V = 0$ $\quad (\nabla^2 V = -\rho/\varepsilon_0)$

Cartesian coordinates:

$$\frac{d^2V}{dx^2} + \frac{d^2V}{dy^2} + \frac{d^2V}{dz^2} = 0$$

for 1 dimensional

$$\frac{d^2V}{dx^2} = 0 \Rightarrow V(x) = mx + b$$

$$V(x) = \frac{V(x+a) + V(x-a)}{2}$$ averaging instruction

Solution of Laplace's equation can't have local minima / local maxima

Earnshaw's Theorm: A charged particle can't be held in stable equilibrium by electrostatic forces alone

Polarization: induced dipole moment per unit volume (P)

$\vec{p} = \alpha \vec{E}$ ($\alpha \rightarrow$ polarizability) — property of material; how easy / difficult it is to produce the dipoles

$p \rightarrow$ induced dipole moment $\vec{E} \rightarrow$ Applied electric field

Inducing dipoles when kept in external Electric Field → Polarization

4 Types:

a) Atomic / Electronic

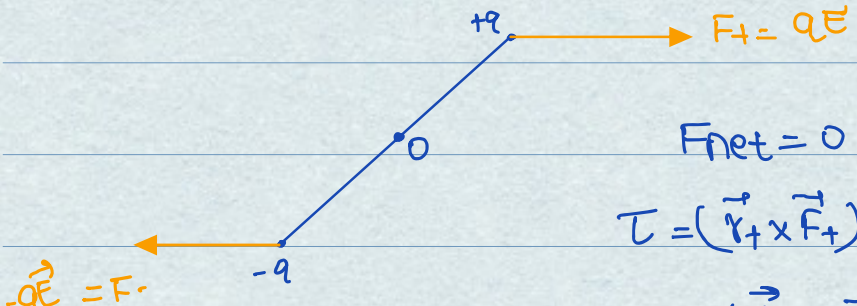
b) Ionic

c) Orientational

d) Space charge

Dielectrics: materials which can't conduct charges but can store charges.

Dipole in electric field:



$F_{net} = 0$ (cancels)

$\tau = (\vec{r}_+ \times \vec{F}_+) + (\vec{r}_- \times \vec{F}_-)$

$= \left(\frac{\vec{d}}{2} \times q\vec{E}\right) + \left(-\frac{\vec{d}}{2} \times -q\vec{E}\right)$

$= q\vec{d} \times \vec{E} = \boxed{\vec{p} \times \vec{E}}$

If field is not uniform,

$\vec{F}_{net} = q[\vec{E}_+ - \vec{E}_-] = q\Delta\vec{E}$

$\Delta\vec{E}_x = (\nabla E_x) \cdot \vec{d} \Rightarrow \Delta\vec{E} = (\vec{d} \cdot \nabla)\vec{E}$

$\boxed{\vec{F} = (\vec{p} \cdot \nabla)\vec{E}} = p_x \frac{dE_x}{dx} + p_y \frac{dE_y}{dy} + p_z \frac{dE_z}{dz}$

Field due to polarized object:

$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{\vec{p} \cdot \hat{R}}{R^2}$

$= \frac{1}{4\pi\varepsilon_0} \int \frac{\hat{R}\vec{P}(\vec{r}')}{R^2} d\tau'$

$\vec{r}'$ → from 0 to dipole

$\vec{r}$ → from 0 to point

We know $\boxed{\nabla'(1/R) = \frac{\hat{R}}{R^2}}$

$\nabla = \frac{d}{dx} + \frac{d}{dy} + \frac{d}{dz}$

$\Rightarrow \nabla' = \frac{d}{dx'} + \frac{d}{dy'} + \frac{d}{dz'}$

$R^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_V \vec{P}\cdot\nabla'\left(\frac{1}{R}\right)d\tau'$$

$\phi \rightarrow$ scalar

$$\vec{\nabla}(\phi\vec{E}) = \vec{\nabla}(\phi)\,\vec{E} + \phi\vec{\nabla}\cdot\vec{E}$$

$$\Rightarrow V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left[\int_V \nabla'\left(\frac{\vec{P}}{R}\right)d\tau' - \int_V \frac{1}{R}(\nabla'\cdot\vec{P})\,d\tau'\right]$$

using divergence theorm $\left[\int_V \nabla\vec{F}\cdot dv = \int_S \vec{F}\cdot\vec{n}\,d\vec{s}\right]$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\oint_S \frac{1}{R}\vec{P}\cdot d\vec{a}' - \frac{1}{4\pi\varepsilon_0}\int_V \frac{1}{R}(\nabla'\cdot\vec{P})\,d\tau'$$

↓ potential produced by a surface charge density σ_b

↓ potential produced by a volume charge density ρ_b

$\sigma_b = \vec{P}\cdot\hat{n}$

$\rho_b = -\vec{\nabla}\cdot\vec{P}$

$$V = \frac{1}{4\pi\varepsilon_0}\oint_S \frac{\sigma_b}{R}\,d\vec{a}' + \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho_b}{R}\,d\tau'$$

we can find just field due to those bound charges. (uniformly polarized: $\rho_b = 0$)

Gauss law in presence of dielectric

free charge not a result of polarization

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} = \frac{\rho_f + \rho_b}{\varepsilon_0} \Rightarrow \boxed{\varepsilon_0 \vec{\nabla}\cdot\vec{E} = -\vec{\nabla}\cdot\vec{P} + \rho_f}$$

(ρ_f: any free charge; ρ_b: bound charge)

$\vec{E}$: total field, not just that portion due to polarization

$$\boxed{\vec{\nabla}\cdot(\varepsilon_0\vec{E} + \vec{P}) = \rho_f}$$

$$\boxed{\vec{D} = \varepsilon_0\vec{E} + \vec{P}} \quad \text{electric displacement}$$

$$\Rightarrow \boxed{\vec{\nabla}\cdot\vec{D} = \rho_f} \quad \text{differential form}$$

(or) $$\boxed{\oint \vec{D}\cdot d\vec{a} = (Q_f)_{enclosed}} \quad \text{integral form}$$

$(Q_f)_{enclosed}$: total free charge enclosed (not bound charges)

Linear Dielectrics:

Polarization is proportional to the field provided the field is not too strong

$$\boxed{\vec{P} = \varepsilon_0 \chi_e \vec{E}}$$ $\chi_e \rightarrow$ dielectric susceptibility

Materials that accept this equation are the linear dielectrics in linear media

$$\boxed{\vec{D} = \varepsilon_0\vec{E} + \vec{P} = \varepsilon_0(1+\chi_e)\vec{E}}$$

$$\vec{D} = \varepsilon\vec{E} \qquad \boxed{\varepsilon = \varepsilon_0(1+\chi_e)} \rightarrow \text{permittivity of the material}$$

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e$$ → relative permittivity or dielectric constant (material property)

Energy in dielectrics

$$W = \frac{\varepsilon_0}{2} \int_{\text{all space}} |\vec{E}|^2 dV \qquad W = \frac{1}{2} \int \phi \rho \, dV$$

$$\Delta W = \frac{1}{2} \int (\Delta \rho_f) V \, d\tau \qquad \vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\Rightarrow \Delta \rho_f = \vec{\nabla} \cdot (\Delta \vec{D})$$

$$\Delta W = \frac{1}{2} \int [\vec{\nabla} \cdot (\Delta \vec{D})] V \, d\tau$$

using product rule: $\nabla(\phi \vec{E}) = \vec{\nabla}\phi \cdot \vec{E} + \phi \vec{\nabla} \cdot \vec{E}$

$$\vec{\nabla} [(\Delta \vec{D}) V] = \vec{\nabla} \cdot (\Delta \vec{D}) V + \Delta \vec{D} \cdot (\vec{\nabla} V)$$

$$\Delta W = \frac{1}{2} \int \vec{\nabla} \cdot [(\Delta \vec{D}) V] \, d\tau + \frac{1}{2} \int (\Delta \vec{D}) \cdot \vec{E} \, d\tau$$

$$\Delta W = \frac{1}{2} \oint (\Delta \vec{D}) V \cdot \vec{da} + \frac{1}{2} \int (\Delta \vec{D}) \cdot \vec{E} \, d\tau$$

vanishes if we integrate over all space

$$\Delta W = \frac{1}{2} \int_{\text{all space}} (\Delta \vec{D} \cdot \vec{E}) \, d\tau \qquad W = \frac{1}{2} \int_{\text{all space}} \vec{D} \cdot \vec{E} \, d\tau$$

Linear dielectric $\Rightarrow \vec{D} = \varepsilon \vec{E}$

$$\frac{1}{2}\Delta(\vec{D}\cdot\vec{E}) = \frac{1}{2}\Delta(\varepsilon E^2) = \frac{1}{2}\varepsilon(\Delta\vec{E})\cdot\vec{E} = \frac{1}{2}(\Delta\vec{B})\cdot\vec{E}$$

$$\Rightarrow \Delta w = \Delta\left(\frac{1}{2}\int \vec{D}\cdot\vec{E}\, d\tau\right)$$

$$\Rightarrow \boxed{w = \frac{1}{2}\int_{\text{all space}} \vec{D}\cdot\vec{E}\, d\tau}$$

- There is no Columb's law for D
- curl $E = 0$ but curl $D \neq 0$

$$\vec{\nabla}\times D = \vec{\nabla}\times P$$

Boundary conditions:

$$D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f$$

$$D''_{above} - D''_{below} = P''_{above} - P''_{below}$$

$$E^{\perp}_{above} - E^{\perp}_{below} = \frac{\sigma}{\varepsilon_0}$$

$$E''_{above} - E''_{below} = 0$$

Quadrupole moment tensor

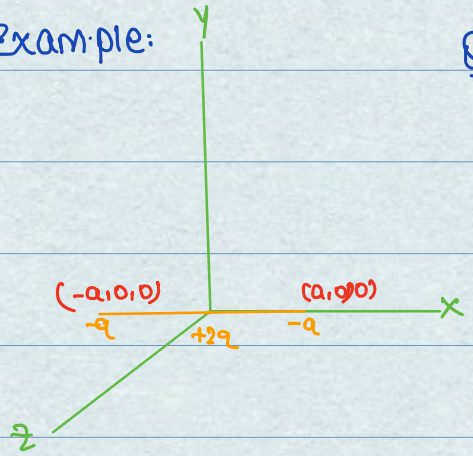
$i,j = x,y,z \qquad Q_{ij} = \sum_k q_k (3 r_{ki} r_{kj} - r_k^2 \delta_{ij})$

$$V = \frac{1}{8\pi\varepsilon_0 r^3} \sum_i \sum_j Q_{ij}\, \hat{r}_i\, \hat{r}_j$$

$\Rightarrow$

$$V_{Quad} = \frac{1}{4\pi\varepsilon_0 R^3}\frac{1}{2R^2}\begin{bmatrix} x & y & z \end{bmatrix}\begin{bmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{yx} & Q_{yy} & Q_{yz} \\ Q_{zx} & Q_{zy} & Q_{zz} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Example:



$$Q_{xx} = \sum_k q_k (3r_x^2 - r_k^2)$$

$$= -q(3a^2 - a^2) \times 2$$

$$= -4qa^2$$

$$Q_{yy} = \sum_k q_k (3r_y^2 - r_k^2)$$

$$= -qa^2 \times 2 = -2qa^2 = Q_{zz}$$

$$Q = \begin{bmatrix} -4qa^2 & 0 & 0 \\ 0 & -2qa2 & 0 \\ 0 & 0 & -2qa^2 \end{bmatrix}$$

$$V = \frac{1}{8\pi\varepsilon_0 R^5} \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} -4qa^2 & 0 & 0 \\ 0 & -2qa^2 & 0 \\ 0 & 0 & -2qa^2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \frac{1}{8\pi\varepsilon_0 R^5} \begin{bmatrix} -4qa^2x & -2qa^2y & -2qa^2z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow V(\vec{R}) = \frac{1}{8\pi\varepsilon_0 R^5} \left[-4qa^2x^2 - 2qa^2y^2 - 2qa^2z^2 \right]$$

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -\vec{\nabla}\left(\varepsilon_0 \frac{\chi_e}{\varepsilon} \vec{D}\right) = -\left(\frac{\chi_e}{1+\chi_e}\right)\rho_f$$

For linear homogeneous dielectric: $\rho_b \propto \rho_f$

MAGNETOSTATICS

Stationary charges produces only electric field around it. But a moving charge generates a magnetic field as well.

When a current flows through a straight conductor, magnetic field is generated circumferentially

$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi}\int \frac{d\vec{l}\times\hat{R}}{R^2}$$ Biot-Savert's law experimentally arrived

charge moving with constant velocity

$\Rightarrow$ steady currents $\Rightarrow$ constant magnetic fields

$\Rightarrow$ magnetostatics

$$\vec{F}_{magnetic} = q(\vec{v}\times\vec{B})$$ ($\vec{v}$: velocity) $\rightarrow$ Lorentz force law

in presence of both $\vec{E}$ and $\vec{B}$

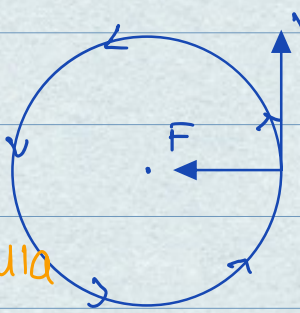
$$\vec{F} = q(\vec{E}+\vec{v}\times\vec{B})$$ Lorentz force

motion of charged particle in magnetic field is circular

$$qvB = \frac{mv^2}{R}$$

$$\Rightarrow \boxed{p = qBr}$$

cyclotron formula



cyclotron motion

If there's a parallel component of v along with $\vec{B}$, particle moves in a helix.

Magnetic forces don't do any work.
They can alter the direction of motion, but can't change its speed

Currents: charge per unit time passing a given point 1 Ampere = 1 Coulomb / second

Line charge (λ): $\vec{F} = \int (\vec{v} \times \vec{B}) \lambda \, dl = \int (\vec{I} \times \vec{B}) \, dl$

$\vec{I} = \lambda \vec{v}$ $= \int I (d\vec{l} \times \vec{B})$

Surface charge (σ): $\vec{F} = \int (\vec{v} \times \vec{B}) \sigma \, da = \int (\vec{K} \times \vec{B}) \, da$

$\vec{K}$ = surface current density $= \sigma \vec{v} = \frac{d\vec{I}}{dl}$

volume charge (ρ): $\vec{F} = \int (\vec{v} \times \vec{B}) \rho \, dv = \int (\vec{J} \times \vec{B}) \, dv$

$\vec{J}$ = current per unit area perpendicular to the flow $= \rho \vec{v} = \frac{d\vec{I}}{da_\perp}$

current crossing a surface $\Rightarrow \boxed{I = \int_S \vec{J} \cdot d\vec{a}}$

$\boxed{\oint_S \vec{J} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{J}) \, dv}$ → total charge per unit time leaving a volume V

gauss law

what comes out of the surface should come at the expense of the remaining inside ← charge conservation

$$\int_V (\vec{\nabla} \cdot \vec{J}) \, dv = -\frac{d}{dt} \int_V \rho \, dv = -\int_V \frac{\partial \rho}{\partial t} \, dv$$

decrease in charge in volume

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}}$$ → continuity equation

conservation of charge

true for any volume

magnetostatics $\Rightarrow \frac{\partial \rho}{\partial t} = 0 \Rightarrow \boxed{\vec{\nabla} \cdot \vec{J} = 0}$

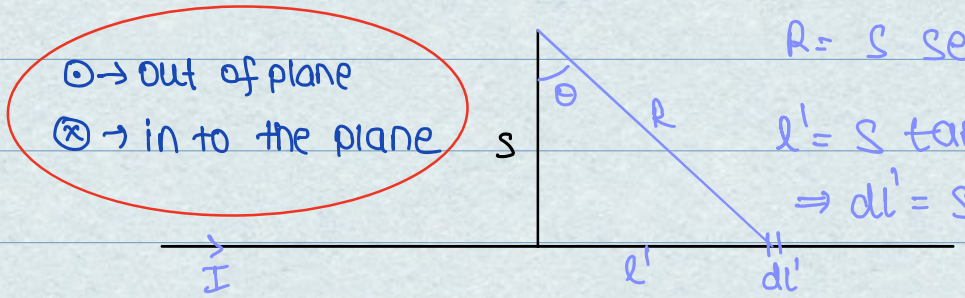
Biot Savert's law:

$$\boxed{\vec{B}(\vec{r})_{\text{steady line current}} = \frac{\mu_0}{4\pi} I \int \frac{d\vec{l} \times \hat{R}}{R^2}}$$

$\boxed{\mu_0 = 4\pi \times 10^{-7} \, N/A^2 = \text{permeability of free space}}$

Integration is along current direction ($\vec{dl}$)

Long straight wire, current I. Field at distance s from wire



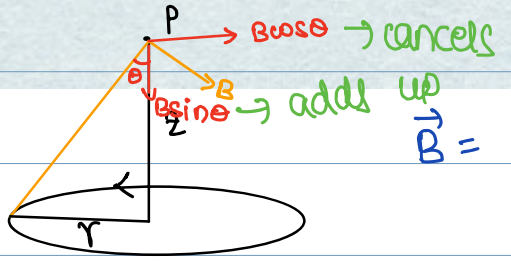
$R = s\sec\theta$

$l' = s\tan\theta$

$\Rightarrow dl' = s\sec^2\theta\, d\theta$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\int_{\theta_1}^{\theta_2} \frac{s\sec^2\theta\, d\theta \cdot s\sec\theta\cos\theta}{s^3\sec^3\theta} = \frac{\mu_0 I}{4\pi s}(\sin\theta_1 - \sin\theta_2)$$

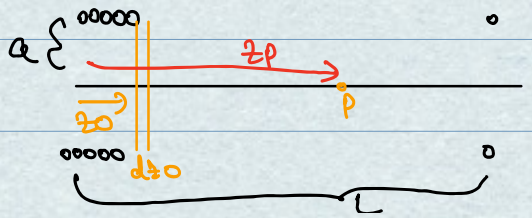
$$\vec{B} = \frac{\mu_0 I}{2\pi s} \quad (\text{out of plane})$$



$$\vec{B} = \frac{\mu_0 I}{4\pi}\int \frac{\vec{dl}\cdot\hat{R}\sin\theta}{R^2}$$

$$= \frac{\mu_0 I}{4\pi 2}\frac{r\times 2\pi r}{(\sqrt{r^2+z^2})^3} = \frac{\mu_0 I r^2}{2(r^2+z^2)^{3/2}} \quad (\text{towards the ring})$$

Solenoid; radius a; n → turns per unit length



$$dB = \frac{\mu_0 I a^2 n\, dz_0}{2[a^2+(z_p-z_0)^2]^{3/2}}$$

$$\Rightarrow B = \frac{\mu_0 I a^2 n}{2a^3} \int \frac{d\theta}{\sec\theta}$$

$$= \frac{\mu_0 I n}{2} \left| \sin\theta \right|_{\theta_1}^{\theta_2}$$

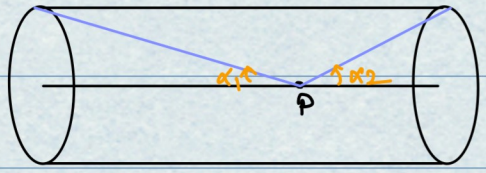
$$= \frac{\mu_0 n I}{2} \left[\frac{z_p - L}{\sqrt{(z_p - L)^2 + a^2}} - \frac{z_p}{\sqrt{z_p^2 + a^2}} \right]$$

$z_p - z_0 = a\tan\theta$ $\quad z_0 = 0$ to L

$dz_0 = a\sec^2\theta\, d\theta$

$\frac{a\sec^2\theta\, d\theta}{a^3 \sec^3\theta}$

$\tan\theta_1 = \frac{z_p}{a}$ $\quad \tan\theta_2 = \frac{z_p - L}{a}$



$$\Rightarrow B = \frac{\mu_0 n I}{2} [\cos\alpha_1 + \cos\alpha_2]$$

Infinite length solenoid $\Rightarrow B = \mu_0 n I$

Divergence and curl of $\vec{B}$.

infinite wire: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



independent of radius

$\Rightarrow$ B decreases as circumference increases

Bundle of straight wires: $\boxed{\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}}$ ← ampere's law in integral form

$I_{enc} = \int \vec{J} \cdot d\vec{a}$ $\quad$ J → volume current density

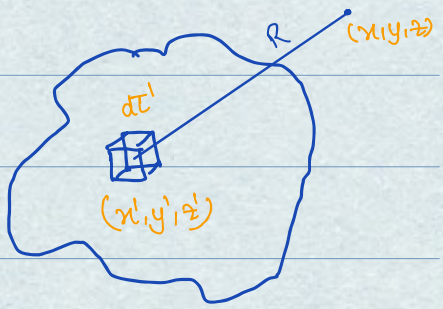
(stokes theorm) $\int (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \mu_0 \int \vec{J} \cdot d\vec{a}$

$\Rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}}$ → valid only if wire is infinitely long

shows that magnetic field is rotational

→ ampere's law in differential form

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{J(\vec{r}')\, x\hat{R}}{R^2}\, d\tau'$$



B is a function of (x,y,z)

J is a function of (x',y',z')

$$\vec{R} = (x-x')\hat{e}_x + (y-y')\hat{e}_y + (z-z')\hat{e}_z$$

$$d\tau' = dx'\,dy'\,dz'$$

$$\vec{\nabla}\cdot\vec{B} = \frac{\mu_0}{4\pi} \int \vec{\nabla}\cdot\left(\vec{J} \times \frac{\hat{R}}{R^2}\right) d\tau'$$

$$\boxed{\vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{A}) - \vec{A}\cdot(\vec{\nabla}\times\vec{B})}$$

$\vec{\nabla}\times\vec{J} = 0$ (J doesnt depend on unprimed coordinates)

$$\boxed{\vec{\nabla}\times\left(\frac{\hat{R}}{R^2}\right) = 0} \Rightarrow \boxed{\vec{\nabla}\cdot\vec{B} = 0}$$

→ proves that magnetic field is solenoidal

$$\vec{\nabla}\times\vec{B} = \frac{\mu_0}{4\pi} \int \vec{\nabla}\times\left(\vec{J}\times\frac{\hat{R}}{R^2}\right) d\tau'$$

$$\boxed{\vec{\nabla}\times(\vec{A}\times\vec{B}) = \vec{A}(\vec{\nabla}\cdot\vec{B}) - \vec{B}(\vec{\nabla}\cdot\vec{A})}$$

$$\vec{\nabla}\cdot\vec{J}=0\,,\quad \vec{\nabla}\cdot\left(\frac{\hat{R}}{R^2}\right)=4\pi\,\delta^3(\vec{R})$$

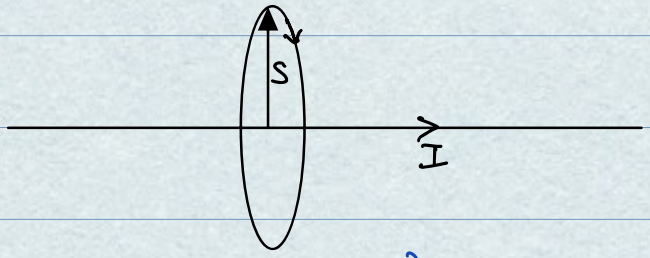
$$\Rightarrow \vec{\nabla}\times\vec{B}=\frac{\mu_0}{4\pi}\int J(\bar{r}')\,4\pi\,\delta^3(\bar{r}-\bar{r}')\,d\tau'$$

$$=\mu_0 J(\bar{r})$$

Ampere's law in diff. form

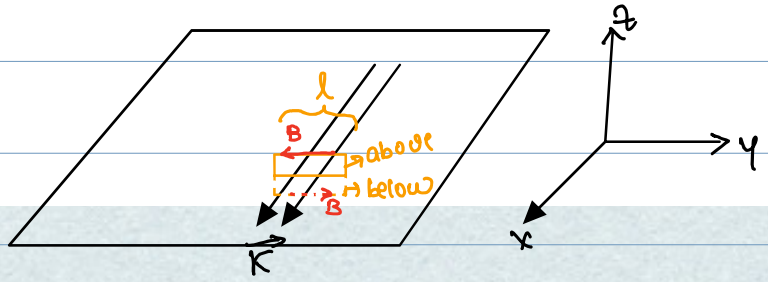
$$\int(\vec{\nabla}\times\vec{B})\cdot da=\int\vec{B}\cdot\vec{dl}=\mu_0\int\vec{J}\cdot\vec{da}=\mu_0 I_{enc}$$

Ampere's law in integral form



$$\oint\vec{B}\cdot\vec{dl}=B\oint\vec{dl}=B\cdot 2\pi s=\mu_0 I_{enc}=\mu_0 I$$

$$\Rightarrow B=\frac{\mu_0 I}{2\pi s}$$



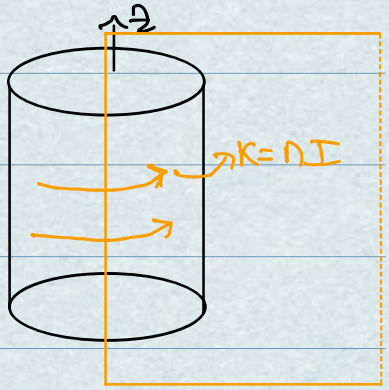
$\bar{K}=K\hat{e}_x$

B can't have an x-component since $B\perp K$

B can't have a z component due to symmetric cancellation from +y and -y elements

$$2B\cancel{\ell} = \mu_0 K\cancel{\ell} \Rightarrow \frac{\mu_0 K}{2} = B$$

Long solenoid: radius R, current I, n→turns per unit length ⇒ similar to cylinder sheet



$$B_\phi 2\pi s = \mu_0 I_{enc} = 0 \Rightarrow B_\phi = 0$$

$$B_r = 0$$

$$B_z : B_z \cdot \ell - 0 = \mu_0 (nI\ell)$$

$$\Rightarrow B_z = \mu_0 nI\,\hat{z}$$

Field is zero outside solenoid (along radial direction)

Maxwell's equations:

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \vec{\nabla}\cdot\vec{B} = 0 \qquad \vec{\nabla}\times\vec{E} = 0 \qquad \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$$

Magnetic field are produced by electric charges in motion and it takes another moving charge to feel the magnetic field

Magnetic vector potential:

$\vec{B}(\vec{r}) = \vec{\nabla} \times \vec{A}(\vec{r})$ B = curl of some vector A

$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}(\vec{r})) = 0$ (divergence of curl)

$\vec{A}(\vec{r})$ → magnetic vector potential

$\vec{A}$ units: volt-second/meter

$$B = \frac{\mu_0 I}{4\pi} \int d\vec{l}' \times \left\{ -\text{grad}\left(\frac{1}{r}\right) \right\}$$

$$\nabla\left(\frac{1}{r}\right) = -\frac{\hat{r}}{r^2}$$

$$\text{curl}(S\bar{A}) = S\,\text{curl}\,\bar{A} - \bar{A}\,\text{grad}\,S$$

$$\Rightarrow -\bar{A}\,\text{grad}\,S = \text{curl}(S\bar{A}) - S\,\text{curl}(\bar{A})$$

$$B = \frac{\mu_0 I}{4\pi} \int \left\{ \text{curl}\left(\frac{d\bar{l}'}{r}\right) - \frac{1}{r}\,\text{curl}(d\bar{l}) \right\}$$

Second term in integral is zero because $d\vec{l}'$ doesn't depend on coordinates of field point.

$$B = \text{curl}\left(\frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}}{r} \right)$$

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi} \oint_{\text{line}} \frac{I\,d\vec{l}'}{r} = \frac{\mu_0}{4\pi} \int_{v'} \frac{\vec{J}\,dv'}{r} = \frac{\mu_0}{4\pi} \int_{a'} \frac{\vec{K}\,da}{r}$$

A satisfies the poisson's equation:

$$\nabla^2 A = \frac{\mu_0}{4\pi}\int \nabla^2\left(\frac{\vec{J}'}{r}\right)d\tau' = \frac{\mu_0}{4\pi}\int J' \underbrace{\nabla^2\left(\frac{1}{r}\right)}_{-4\pi} d\tau' = -\mu_0\vec{J}'$$

$$\boxed{\nabla^2 A = -\mu_0 J'} \qquad \boxed{\int \nabla^2\left(\frac{1}{r}\right)d\tau' = -4\pi}$$

line integral of A around a closed path gives the magnetic flux linked with the area enclosed by the closed path

$$\boxed{\phi_B = \oint \vec{A}\cdot d\vec{l}}$$

Stokes theorem

$$\phi_B = \int_S \vec{B}\cdot d\vec{S} = \int_S (\nabla \times A)\cdot dS = \oint \vec{A}\cdot d\vec{l}$$

Divergence of A is zero or a scalar constant

$$\nabla \times B = \nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A$$

$$\mu_0 J = \nabla(\nabla \cdot A) + \mu_0 J$$

$$\Rightarrow \boxed{\nabla(\nabla \cdot A) = 0} \Rightarrow \nabla \cdot A \text{ is zero / forms a uniform scalar field with constant value}$$

We can add to magnetic potential the gradient of any scalar whose curl vanishes with no effect on B.

We use this freedom to eliminate divergence of A and make it 0. ($\lambda \rightarrow$ scalar function)

$\vec{\nabla}\cdot\vec{A}=0$ $\qquad$ $\vec{\nabla}\times\vec{A}=\vec{\nabla}\times\vec{A}_0=B$

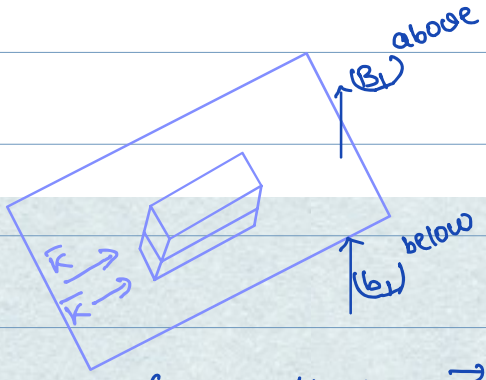
$\boxed{\vec{A}=\vec{A}_0+\vec{\nabla}\lambda} \Rightarrow \vec{\nabla}\cdot\vec{A}=\vec{\nabla}\cdot\vec{A}_0+\nabla^2\lambda$

$\Rightarrow \nabla^2\lambda=-\vec{\nabla}\cdot\vec{A}_0$

(Similar to poisson's equation $\nabla^2 V=-\rho/\varepsilon_0$. It's always possible to make vector potential divergenceless) $\Rightarrow$ Non-uniqueness of vector potential

Boundary conditions:

$\boxed{(B_\perp)_{above}=(B_\perp)_{below}}$



$\oint \vec{B}\cdot d\vec{l}=\mu_0 I_{enc}$

$\Rightarrow B''_{above}-B''_{below}=0$ $\qquad$ (loop // to $\vec{K}$)

$B''_{above}-B''_{below}=\mu_0(\vec{K})$ $\qquad$ (loop $\perp$ to $\vec{K}$)

($\hat{n} \rightarrow$ unit vector $\perp$ to surface)

Magnetic vector potential is continuous

$$\vec{A}_{above} = \vec{A}_{below}$$

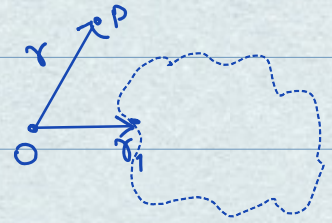
$\vec{\nabla}.\vec{A}=0 \Rightarrow$ guarentees normal component is continuous

$\vec{\nabla}\times\vec{A}=\vec{B} \Rightarrow$ in the form $\oint \vec{A}.d\vec{l} = \oint \vec{B}.d\vec{a} = \emptyset$

$\Rightarrow$ Tangential components are continuous

Multipole expansion of the vector potential for line currents:

$$\frac{1}{R} = \frac{1}{(r^2+(r')^2-2rr'\cos\theta)^{1/2}}$$



$$A(\vec{r}) = \frac{\mu_0 I}{4\pi}\left[\frac{1}{r}\oint d\vec{l}' + \frac{1}{r^2}\oint r'\cos\theta' d\vec{l}' + \frac{1}{r^3}\oint \frac{(r')^2}{2}(3\cos^2\theta'-1)\,d\vec{l}' + \cdots\right]$$

$\oint d\vec{l}' = 0$ Monopole term $= 0$

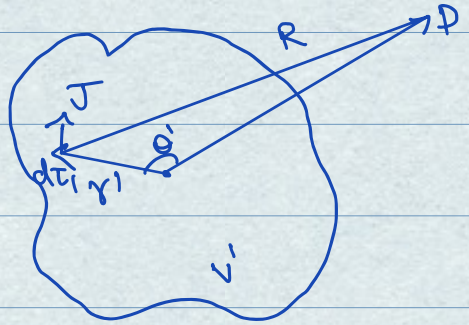
dipole term: $\frac{\mu_0 I}{4\pi r^2}\oint r'\cos\theta' d\vec{l}' = \frac{\mu_0 I}{4\pi r^2}\oint(\hat{r}.\vec{r}')\,d\vec{l}'$

$$\oint (\hat{r}\cdot\vec{r}')\,d\vec{\tau}' = -\hat{r}\times\int d\vec{a}'$$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2} \qquad \vec{m} = I\int d\vec{a} = I\vec{a}$$

For volume currents:

electrostatics: $\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho(\vec{r}')\,d\tau'}{R}$



Here: $\frac{\mu_0}{4\pi}\oint_{V'}\frac{\vec{J}(\vec{r}')\,d\tau'}{R}$

$$\frac{1}{R} = \frac{1}{(r^2+(r')^2-2rr'\cos\theta)^{1/2}}$$

$$\Rightarrow A = \frac{\mu_0}{4\pi r}\int_{V'}J(\vec{r}')\,d\tau' + \frac{\mu_0}{4\pi r^2}\int J(\vec{r}')(\hat{r}\cdot\vec{r}')\,d\tau'$$

$$+\frac{\mu_0}{4\pi r^3}\int\frac{1}{2}J(\vec{r}')\left[3(\hat{r}\cdot\vec{r}')^2 - r'^2\right]d\tau' + \dots$$

$$\int_V \vec{J}(\vec{r}')\,d\tau' = \sum_i I_i\oint_{C_i}d\vec{l}_i = 0$$ (total vector displacement around a closed loop is 0)

$$\vec{A}_{monopole} = 0$$

$$\vec{A}_{dipole}(\vec{r}) = \frac{\mu_0}{4\pi r^2}\int_{V'}\vec{J}(\vec{r}')(\hat{r}\cdot\vec{r}')\,d\tau'$$

Filamentry currents: $J d\tau = I d\vec{l}'$

$$\oint (\vec{C}\cdot\vec{r})\,dl = \left\{\frac{1}{2}\int_C (\vec{r}\times d\vec{l})\right\}\times\vec{C}$$

C → constant vector

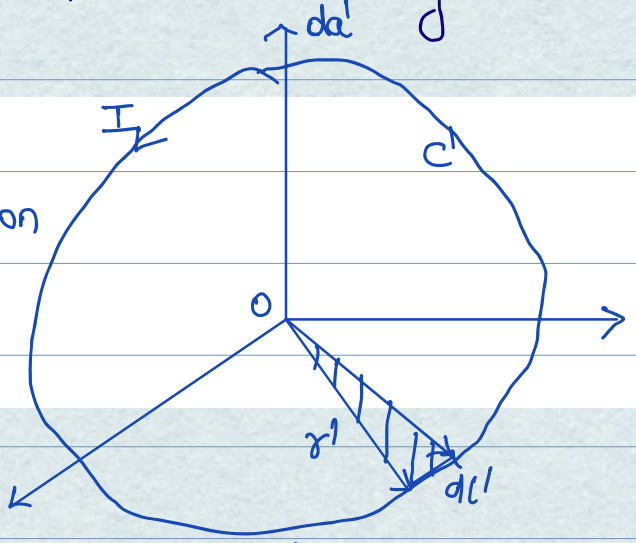
r → any position vector

$$\boxed{m = \frac{1}{2}\int_{V'}\left[\vec{r}'\times\vec{J}(\vec{r}')\right]d\tau'}$$

$$\boxed{m = \frac{I}{2}\int_C \vec{r}'\times d\vec{l}'}$$

$$\boxed{\vec{A_D}(\vec{r}) = \frac{\mu_0}{4\pi r^2}(\vec{m}\times\hat{r})}$$

Dipole moment due to plane filamentary current:



$\frac{1}{2}(\vec{r}'\times d\vec{l}')$ = area of shaded region $= \vec{da}'$

$\Rightarrow I/2 \int_C \vec{r}'\times d\vec{l}'$

$= I\int d\vec{a} = I\vec{S}$

$\Rightarrow \boxed{\vec{m} = I\vec{S} = IS\hat{n}}$

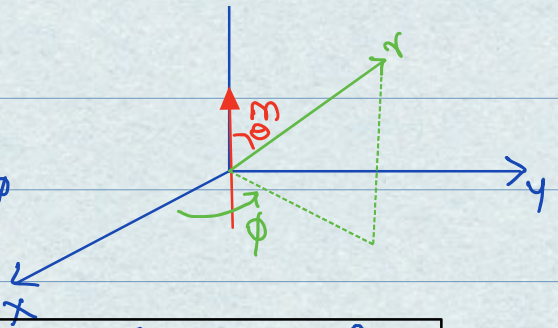
$\vec{da}'$ → ⊥ to plane

Electrostatics: $\vec{p} = q\vec{d}$



$$\vec{A}_{dipole} = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2}$$

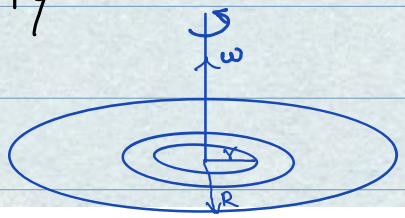
$$= \frac{\mu_0}{4\pi}\frac{m\sin\theta}{r^2}\hat{e}_\phi$$



$$\vec{B}_{dipole} = \vec{\nabla}\times\vec{A} = \frac{\mu_0 m}{4\pi r^3}(2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\theta)$$

$$\vec{B} = \frac{\mu_0 m}{4\pi r^3}[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}]$$

Non conducting disc, radius R, surface charge density σ, $\omega \rightarrow$ angular velocity



$$dQ = \sigma(2\pi r\,dr)$$

$$dI = \frac{dQ}{T} = \frac{\sigma 2\pi r\,dr}{2\pi}\cdot\omega = \sigma\omega r\,dr$$

$$\vec{m} = \int dI\,\vec{a} = \int \sigma\omega r\,dr\cdot\pi r^2 = \frac{\sigma\vec{\omega}\pi R^4}{4} \quad (\text{upward})(\hat{\omega})$$

An electron of mass m and charge $-q$ is moving under the influence of a uniform electric field, $\vec{E} = E_0\hat{y}$ and a uniform magnetic field, $\vec{B} = B_0\hat{z}$. The initial position of the electron is the origin (0, 0,0) and the initial velocity is $\vec{v} = v_0\hat{x}$.
(i) Write the equations of motion. (ii) Determine the trajectory $\vec{r}(t)$ of the electron at time $t > 0$. **[4 Marks]**

let $v = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ $\qquad v_z = 0 = a_z$

$$a_x = -\frac{qB_0v_y}{m} \qquad a_y = \frac{qB_0v_x - qE_0}{m}$$

$$\frac{dv_x}{dt} = -\frac{qB_0v_y}{m} \qquad \frac{dv_y}{dt} = \frac{qB_0v_x - qE_0}{m}$$

$$\frac{d^2v_x}{dt^2} = -\frac{qB_0}{m}\left[\frac{qB_0v_x - qE_0}{m}\right] \qquad \frac{qB_0}{m} = a \qquad \frac{qE_0}{m} = b$$

$$\frac{d^2v_x}{dt^2} = -a^2v_x + ab$$

$$v_x = A\sin(at+\delta) + \frac{b}{a} \qquad A = v_0 - E_0/B_0$$

$$v_y = -A\cos\left(at + \frac{\pi}{2}\right)$$

$$v_{E\times B} = (v_0 - E_0/B_0)\cos\left(\frac{qB_0}{m}t\right) + E_0/B_0 \qquad v_{\bar{E}} = -(v_0 + E_0/B_0)\sin\left(\frac{qB_0t}{m}\right)$$

$$x(t) = \frac{v_0 - E_0/B_0}{qB_0/m}\sin\left(\frac{qB_0}{m}t\right) + \frac{E_0t}{B_0} \qquad (\bar{E}\times\bar{B})$$

$$y(t) = \frac{v_0 - E_0/B_0}{qB_0/m}\cos\left(\frac{qB_0}{m}t\right) - \frac{v_0 - E_0/B_0}{qB_0/m} \qquad (\bar{E})$$